Trigonometric Systems, Chebyshev Polynomials, and Tridiagonal Determinants

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§1.1 Constraints in Trigonometric Systems

Consider the original problem

$$\sin\alpha+\sin\beta=\frac{1}{2},\qquad \cos\alpha-\cos\beta=\frac{1}{3}.$$

The desired quantity is $\cos\alpha\cos\beta$. Introducing four unknowns $a=\sin\alpha$, $b=\sin\beta$, $c=\cos\alpha$, $d=\cos\beta$ gives the two linear equations

$$a+b=\frac{1}{2},\qquad c-d=\frac{1}{3},$$

but they cannot be solved independently: one must retain the circle equations

$$a^2+c^2=1,\qquad b^2+d^2=1.$$

A determinant argument treating a,b,c,d as free coordinates would introduce spurious solutions. The trigonometric variables must instead be linked through the same pair of angles; their circle constraints are part of the algebraic system.

The right substitution uses the shared geometry of the angles:

$$S=\frac{\alpha+\beta}{2},\qquad D=\frac{\alpha-\beta}{2}.$$

Sum-to-product identities give

$$2\sin S\cos D=\frac{1}{2},\qquad -2\sin S\sin D=\frac{1}{3}.$$

Thus

$$\sin S\cos D=\frac{1}{4},\qquad \sin S\sin D=-\frac{1}{6}.$$

Without dividing by any potentially zero factor, square and add to obtain

$$\sin^2 S=\frac{1}{16}+\frac{1}{36}=\frac{13}{144}.$$

Consequently

$$\cos(\alpha+\beta)=\cos 2S=1-2\sin^2 S=\frac{59}{72}.$$

To find the difference angle, use

$$\cos 2D=\frac{(\sin S\cos D)^2-(\sin S\sin D)^2}{\sin^2 S}=\frac{1/16-1/36}{13/144}=\frac{5}{13}.$$

Now the product-to-sum identity completes the problem:

$$\boxed{\cos\alpha\cos\beta=\frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}=\frac{1}{2}\left(\frac{59}{72}+\frac{5}{13}\right)=\frac{1127}{1872}.}$$

In contrast, dividing the original equations to get $-\tan D=2/3$ is legitimate here only after proving that the common factor $\sin S\cos D$ is nonzero; the first displayed equation supplies exactly that proof.

§1.2 Rational parametrization and multiple-angle formulas

An alternative solution method for some circle equations is

$$t = \tan\frac{x}{2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}.$$

It comes from the standard stereographic parametrization of the unit circle. The point $(-1, 0)$ is missing for finite real t, so a rationally transformed equation must either handle it separately or impose the appropriate domain. This matters when clearing denominators.

The same circle structure produces multiple-angle identities. De Moivre's formula is

$$(\cos x + i \sin x)^n = \cos nx + i \sin nx.$$

Taking real and imaginary parts of the binomial expansion yields

$$\cos nx = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \binom{n}{2j} \cos^{n-2j} x \sin^{2j} x,$$

$$\sin nx = \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} (-1)^j \binom{n}{2j+1} \cos^{n-2j-1} x \sin^{2j+1} x.$$

These formulas answer the original question about the alternating signs and binomial coefficients: both are forced by powers of i.

The addition identities also give second-order recurrences:

$$\cos nx = 2 \cos x \cos((n-1)x) - \cos((n-2)x),$$

$$\sin nx = 2 \cos x \sin((n-1)x) - \sin((n-2)x).$$

The tangent recurrence

$$\tan nx = \frac{\tan((n-1)x) + \tan x}{1 - \tan((n-1)x) \tan x}$$

holds only where all displayed tangent expressions and the quotient are defined. It is often better to work with sine and cosine when the denominator may vanish.

§1.3 Determinants and Trigonometric Recurrences

Let $A_n(\theta)$ be the $n \times n$ tridiagonal matrix with $2\cos\theta$ on its diagonal and 1 on the two adjacent diagonals:

$$A_n(\theta) = \begin{pmatrix} 2\cos\theta & 1 & & \\ 1 & 2\cos\theta & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 2\cos\theta \end{pmatrix}.$$

With $D_n = \det A_n$ and the empty determinant $D_0 = 1$, expansion along the last row gives

$$D_n = 2\cos\theta \, D_{n-1} - D_{n-2}, \qquad D_1 = 2\cos\theta.$$

Define $U_0(x) = 1$, $U_1(x) = 2x$ and

$$U_n(x) = 2xU_{n-1}(x) - U_{n-2}(x).$$

By matching both initial values and the recurrence,

$$\boxed{D_n(\theta) = U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}} \qquad (\sin\theta \neq 0).$$

The trigonometric quotient extends to $\sin\theta = 0$ by continuity, whereas U_n is a polynomial valid for all x. For example,

$$D_2 = 4\cos^2\theta - 1, \qquad D_3 = 8\cos^3\theta - 4\cos\theta.$$

The recurrence and the two initial values determine the determinant uniquely. It is also easy to incorporate a modified first diagonal entry. Let $A_n(\delta, t)$ have off-diagonal entries 1, diagonal entries $2t$ except for the first entry δ, and write $E_n = \det A_n(\delta, t)$. Cofactor expansion yields

$$E_0 = 1, \quad E_1 = \delta, \qquad E_n = 2tE_{n-1} - E_{n-2}.$$

With the convention $U_{-1}(t) = 0$, the unique solution is

$$\boxed{E_n = \delta U_{n-1}(t) - U_{n-2}(t)} \qquad (n \geq 1).$$

For $n = 1$ this gives δ, and for $n = 2$ it gives $2t\delta - 1$. Changing the boundary entry changes the initial conditions, while the same Chebyshev recurrence governs the interior.

The recurrence and special-function formulas above follow the NIST *Digital Library of Mathematical Functions*, Sections 18.5 and 18.9.[1]

§1.4 Why the algebra and trigonometry agree

The common ingredient is a linear recurrence with fixed coefficients. The sequences $\cos nx$, $\sin nx$, and the tridiagonal determinants D_n all satisfy

$$u_n = 2c\,u_{n-1} - u_{n-2}, \qquad c = \cos x.$$

A second-order recurrence is determined by two initial values, and differences between the sequences come from those values rather than a different governing law.

One can make this structural connection explicit with the companion matrix

$$\begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix} = \begin{pmatrix} 2c & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_{n-1} \\ u_{n-2} \end{pmatrix}.$$

Its eigenvalues are $e^{\pm ix}$ when $c = \cos x$. Powers of these eigenvalues explain the appearance of $\sin nx$ and $\cos nx$, while the determinant recurrence arises from Laplace expansion. This is the precise bridge between the seemingly unrelated pages of trigonometry and linear algebra.

[1] NIST Digital Library of Mathematical Functions, Chapter 18, especially §§18.5 and 18.9.